



AI Shopping Research Platform

High-Fidelity Telemetry, Counter-Balancing, & Live Analytics

A premium social computing research system testing AI shopping assistant autonomy, user trust, and multi-agent collaborative teaming interfaces. Featuring exact counterbalanced group assignment and real-time Knack database behavioral tracking.

SYSTEM ENVIRONMENT & TELEMETRY

Primary Database:	Knack Cloud DB (Table 1 / object_3)
Integration Protocol:	JSON Telemetry Serialization via REST API
Client Architecture:	Expo React Native + Web Optimizations

1. Dual-Study Architecture & Landing Portal

The platform is designed to run two separate behavioral research studies concurrently. Rather than using fragile URL parameters, the entrance point is centralized in a sleek Apple-esque study selector page.

STUDY 1: AUTONOMY

- 2 Counterbalanced Scenarios
- Low Autonomy (Recommend)
- High Autonomy (Auto-Purchase)

STUDY 2: COLLABORATION

- 4 Counterbalanced Scenarios
- Autonomy (Low/High)
- Collaborative Teaming (Yes/No)

Portal Features

- Real-time Counterbalance Meter: A dynamic colored progress segment that shows researchers the distribution of respondents across all available scenarios.
- Auto-Resuming Flow: If a user temporarily closes the page, local storage memory locks their initial assignment. Clicking "Continue Study" brings them instantly back to their active chat session without altering counterbalance counts.
- Wallet Reset Telemetry: When starting a new session, the user's wallet balance automatically resets to \$1,000.00, and previous local chat histories are wiped clean to prevent carry-over effects.

2. Round-Robin Randomization Algorithm

To guarantee exact statistical cell parity (e.g., 50/50 for Study 1, and 25/25/25/25 for Study 2), the application implements a round-robin counterbalance algorithm in `src/services/randomizationService.ts`. Instead of simple `math-random` (which leads to highly unequal groups at small sample sizes), the system follows a deterministic assignment queue:

ALGORITHM FLOW SPECIFICATION (TypeScript)

```
// 1. Fetch historical counts from persistent memory
const counts = JSON.parse(localStorage.getItem('study_counts') || '{"S1_LOW":0,"S1_HIGH":0}');

// 2. Identify the scenario with the lowest respondent count
const scenario = counts.S1_LOW <= counts.S1_HIGH ? 'S1_LOW' : 'S1_HIGH';

// 3. Assign user, increment cell, and lock assignment in persistent session
counts[scenario]++;
localStorage.setItem('study_counts', JSON.stringify(counts));
```

3. Structured Telemetry & Logging Protocol

To maintain compatibility with standard Knack fields, all tracking data is compiled into a high-fidelity JSON payload and stored in Table 1 (object_3), field_23 (Name). This enables complete behavioral tracking of clicks, timing, and purchasing decisions.

Event Classification & Payloads

Event Type	Description	Key Telemetry Fields Tracked
ASSIGNED	Fires on study enter.	study, scenario, autonomyLevel, teaming, description
MESSAGE_SENT	Fires on user text chat.	text, charCount, wordCount, timestamp
MESSAGE_RECEIVED	AI response delivery.	aiMessage, actionType, matchedProductId, matchedPrice
DECISION_ACTION	Final transaction state.	action (CONFIRMED DECLINED AUTO_PURCHASED GUARD_BLOCKED), price, walletAfter

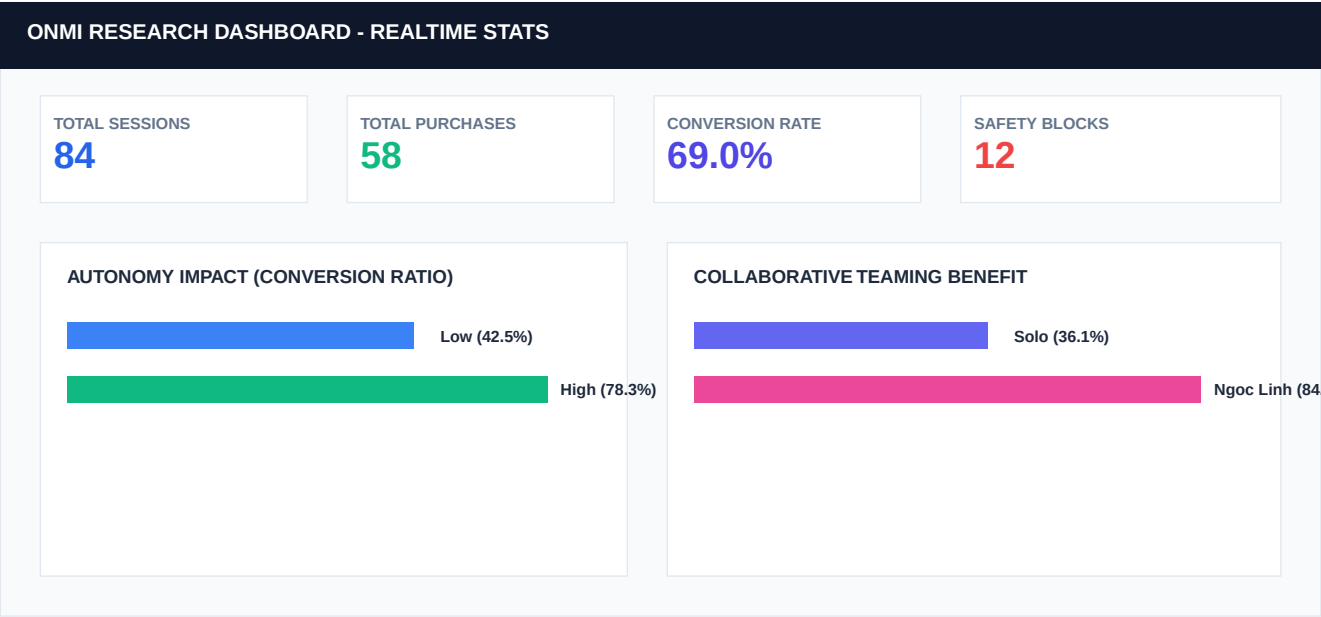
4. Analytical Advantages

By standardizing all actions around a persistent Session ID (e.g. sess_1703112000_a1b2c3d4), researchers gain tremendous advantages during analysis:

- Timeline Reconstruction: Easily order actions chronologically to trace the exact dialog history and decision milestones for each respondent.
- Conversion Funnel Modeling: Measure drop-offs, refusal counts (Alternative Request), and purchase rates based on varying autonomy levels.
- Safety Intercept Analysis: Track how safety budget guard limits prevent High Autonomy agent checkout, showing the limits of agency.

5. Live Analytics & Research Dashboard Guide

A full-featured analytics suite is built directly into the client at /dashboard. It fetches live, raw telemetry from the Knack database and aggregates them instantly into visual indicators and inspectable tables.



Operations & Exporting Instructions

1. Live Refresh: Click the indigo "Refresh" button at the top-right to pull current logs from the Knack REST API instantly. The loading spinner will indicate network progress.
2. Respondent Analysis Grid: The bottom half of the dashboard shows all active sessions. Clicking a row dynamically populates the step-by-step chat history panel on the right.
3. CSV Exporting: Click "Export CSV" to immediately trigger a browser file download of the full, flattened data structure. The system parses the JSON column and compiles it into a clean, comma-delimited Excel-compatible format.

